MIL-C-10888D(AR)
7 OCTOBER 1986
SUPERSEDING
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20 June 1972

MILITARY SPECIFICATION
CANNON, 4.2 INCH MORTAR: M30
AND
MOUNT, 4.2 INCH MORTAR: M24A1

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification supports the acquisition of the 4.2 inch M30 mortar cannon and the 4.2 inch M24A1 mortar ground mount.

1.2 This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Product Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-C-13931 - Cannon, General Specification for
MIL-P-14232 - Parts, Tools and Equipment for Army Material, Packaging of
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1015

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STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1189 - Standard Symbology for Marking Unit Packs, Outer Containers, and Selected Documents

2.1.2 Other Government drawings. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Army Armament, Munitions and Chemical Command

EPL 8401603	- Engineering Parts List for Mount, 4.2 Inch Mortar: M24A1
EPL 11577220	- Engineering Parts List for Cannon, 4.2 Inch Mortar: M30
Master LQAP 8401603	- Master Index of Quality Assurance Provisions for Mount, 4.2 Inch Mortar: M24A1
Master LQAP 11577220	- Master Index of Quality Assurance Provisions for Cannon, 4.2 Inch Mortar: M30
8401603	- Mount, 4.2 Inch Mortar: M24A1
11577220	- Cannon, 4.2 Inch Mortar: M30
P8401840	- Special Packaging Instructions, Mortar 4.2 Inch Cannon M30 on Mount M24A1 W/E
LSPI 8401840	- List of Special Packaging Instructions, Mortar 4.2 Inch Cannon M30 on Mount M24A1 W/E
QAP-APPENDIX-WVA	- General Quality Assurance Provisions
8767017	- Magnetic Particle and Visual Inspection of Thin-walled Cannon Tubes
8769067	- Magnetic Particle Inspection of Cannon and Mount Components

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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2.3 Order of precedence. The applicable contract contains an order of precedence clause wherein specifications are listed last. With regard to this specification, the order of precedence is drawing requirements, followed by the text of this specification, then the other referenced specifications and standards. Nothing in this specification shall supersede applicable laws and regulations unless a specific exception has been obtained.

3. REQUIREMENTS

3.1 General requirements. Cannon, parts thereof; and mounts, components, and parts thereof shall conform to: the applicable general drawing, 11577220 or 8401603, the drawings and specifications listed in the Engineering Parts List EPL 11577220 and EPL 8401603, the requirements of this specification, the provisions of MIL-C-13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer.

3.3 Functioning. All parts shall operate without interference, erratic movement, or malfunction.

3.3.1 Sight mount. The sight mount shall operate smoothly throughout its entire range of movement.

3.3.2 Barrel pin assembly. The barrel pin 7136051 assembly shall operate smoothly.

3.3.3 Elevating mechanism. The elevating mechanism shall function through the entire operating range. With the mortar set up in firing position, backlash at the crank-handle shall not exceed 10 degrees rotation in any position of elevation or traverse.

3.3.3.1 Locking device. The device for locking the elevating mechanism into the supporting tube in either the high or low position shall be positive in action.

3.3.4 Traversing mechanism. The traversing mechanism shall function through the entire range of movement. With the mortar set up in firing position, backlash of the traversing wheel assembly shall not exceed 10 degrees rotation in any position of traverse or elevation.

3.3.5 Shock absorber assemblies. The shock absorber assemblies on the tube and the shock absorber in the elevating standard shall function smoothly.

3.4 Performance.

3.4.1 High pressure resistance. The mortar shall be capable of withstanding a chamber pressure of 19,000 plus or minus 700 pounds per square inch when tested as specified in 4.5.2.9.

3.4.2 Functioning. There shall be no impairment in the functioning of the mortar components and mount during and after the testing specified in 4.5.2.

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3.4.3 Bore enlargement. Bore diameters measured across the lands after firing no more than 10 rounds shall not exceed the before firing diameters by more than 0.003 inches. There shall be no indication of abnormal wear, damage, or deformation of the bore surface.

3.4.4 Material soundness. After completion of testing, all parts shall be free from cracks and fractures and shall comply with any special soundness requirements specified on the applicable drawings including the provisions of this specification.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. First article - Initial Production Test, FA-IPT (see 4.4).
c. Quality conformance inspection (see 4.5).

4.3 First article inspection. First article mortar components and parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 11577220 for the M30 cannon and Master LQAP 8401603 for the M24A1 mount. First article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First article mortar components and parts thereof shall be representative of the production processes to be used during quantity production.

4.4 First Article - Initial Production Test (FA-IPT). When specified in the contract, (see 6.2.1), First Article mortars shall be subjected to Initial Production Testing by the Government to assure compliance with the requirements of this specification and any additional requirements listed in the contract. Initial Production Testing procedures and the sample sizes of First Article mortars and components thereof shall be as specified in the contract. First

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Article mortars and components thereof shall be representative of the production processes to be used during quantity production.

4.5 Quality conformance inspection. Quality conformance inspection is listed in Table I.

TABLE I. Quality conformance inspection.

Category/Characteristic	Requirement paragraph	Inspection paragraph
Examinations		
Inspection provisions	(F11577220 & F8401603)	4.5.1.1
Packaging	5.1, 5.2, 5.3	4.5.1.2
Tests		
Functioning	3.3	4.5.2.1
Sight mount	3.3.1	4.5.2.2
Barrel pin assy (B7136051)	3.3.2	4.5.2.3
Elevating mechanism	3.3.3	4.5.2.4
Traversing mechanism	3.3.4	4.5.2.5
Backlash	3.3.3, 3.3.4	4.5.2.6
Locking device	3.3.3.1	4.5.2.7
Shock absorber	3.3.5	4.5.2.8
Performance	3.4	4.5.2.9
High pressure resistance	3.4.1	4.5.2.9.1
Cannon		4.5.2.9.1.1
Mount		4.5.2.9.1.2
Bore enlargement	3.4.3	4.5.2.9.2
Material soundness	3.4.4	4.5.2.9.3

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Mortars, components, and parts thereof shall be inspected for conformance to requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 11577220 for the M30 cannon and Master LQAP 8401603 for the M24A1 mount, and QAP-APPENDIX-WVA.

4.5.1.2 Packaging. Packaging inspection for the level designated in the contract shall be performed in accordance with the applicable special packaging instructions and MIL-P-14232. Packaging failing to comply with the packaging requirements of Section 5 and the contract shall be cause for rejection.

4.5.2 Tests.

4.5.2.1 Functioning. Each cannon and mount shall be operated to determine smoothness of operation without any interference. Any evidence of binding shall be cause for rejection.

4.5.2.2 Sight mount. Each sight mount shall be rotated throughout its entire operating range. Failure to function or any evidence of binding shall be cause

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for rejection.

4.5.2.3 Barrel pin assy (7136051). The latch of each barrel pin assembly shall be rotated to lock the barrel pin assembly into the coupling bracket. Failure to function or any evidence of binding shall be cause for rejection.

4.5.2.4 Elevating mechanism. The elevating mechanism of each mount shall be moved through its entire range of operation by rotating the elevating handle. Failure to function or any evidence of binding shall be cause for rejection.

4.5.2.5 Traversing mechanism. The traversing mechanism of each mount shall be moved through its entire range of operation by rotating the traversing wheel assembly. Failure to function or any evidence of binding shall be cause for rejection.

4.5.2.6 Backlash. Backlash in the elevating and traversing mechanism of each mount shall be measured under the conditions specified in 3.3.3 and 3.3.4. Backlash in excess of that specified in 3.3.3 and 3.3.4 shall be cause for rejection.

4.5.2.7 Locking device. The functioning of the locking device for the tube elevating mechanism of each mount shall be tested in accordance with the following procedures (see 3.3.3.1):

a. Rotate the cam to withdraw the plungers and then raise the traversing mechanism.
b. Release the cam and continue raising the traversing mechanism until the high position is reached.
c. Rotate the cam to withdraw the plungers, lower the traversing mechanism and check the locking action in the lower position of the traversing mechanism.
d. At the high and low positions, the plungers shall make a complete entrance into the locking holes.

Failure to function as specified shall be cause for rejection.

4.5.2.8 Shock absorbers. Each assembled mortar shall be gymnasticated by manually depressing the muzzle end of the cannon in a downward direction through the full range of all shock absorbers and releasing. The gymnastication shall be performed with the mount in each position specified in Table II. Functioning of the shock absorber assemblies shall be smooth and without any evidence of binding. After each gymnastication, the cannon shall return to its original firing position, specified in Table II. Failure to comply with the requirements shall be cause for rejection.

4.5.2.9 Performance.

4.5.2.9.1 Actual firing. Each cannon and mount shall be proof fired by Government personnel at a proving ground designated by USATECOM.

4.5.2.9.1.1 Cannon firing schedule. Each cannon or spare cannon shall be tested by firing one round of ammunition adjusted to produce a chamber pressure of 19,000 plus or minus 700 pounds per square inch. A proof facility mount may be used if necessary.

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4.5.2.9.1.2 Mount firing schedule. Each mount or spare mount shall be tested by the schedule of Table II, with the mount in the specified traverse and elevation positions. A proof facility cannon may be used if necessary. Whenever possible, cannons and mounts shall be tested together by the schedule of Table II, with the mount in the specified traverse and elevation positions.

TABLE II. Mount firing schedule.

Number of rounds	Chamber pressure (psi)	Position	
		Traverse	Elevation (mils)
1	12,600	Center	1015
1	16,800	Max, right	907
1	19,000	Center	800
1	16,800	Max, left	907

4.5.2.9.2 Bore enlargement. After firing no more than 10 rounds, the bore diameter across the lands of the cannon shall be measured with an air gage, an electronic gage or other approved gage. Measurements shall be recorded. Enlargements exceeding the limit specified in 3.4.3 shall be cause for rejection.

4.5.2.9.3 Material soundness. After firing, each cannon, each mount, and parts thereof shall be examined visually for freedom from cracks, fractures and deformation. The exterior surfaces of the barrel tube shall be magnetic particle inspected in accordance with the procedures and acceptance criteria of 8767017. The bore of the barrel shall be borescoped to assure compliance with the acceptance criteria of 8767017. The barrel cap shall be magnetic particle inspected in accordance with the procedures and acceptance criteria of 8769067, Class III. The bridge cup, baseplate rotater, and baseplate shall be magnetic particle inspected in accordance with the procedures and acceptance criteria of 8769067, Class IV. Failure to comply with the requirements shall be cause for rejection.

4.5.2.9.4 Records. Cannon and mounts that pass the firing tests shall be recorded by the proving ground by entering applicable data on DA Form 2408-4, "Weapon Record Data".

4.5.3 Inspection equipment.

4.5.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquisition, maintenance, and disposition of inspection equipment prescribed in applicable item QAPs, and for all other inspection equipment required to perform inspections prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of commercial inspection equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being measured.

5. PACKAGING

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5.1 Cannon - Levels A, B and C. Preservation, packaging and packing of the cannon shall be in accordance with Special Packaging Instruction P8401840 for the level of protection specified in the contract (see 6.2.1).

5.2 Repair parts.

5.2.1 Levels A, B and C. Repair parts preservation, packaging and packing shall be in accordance with the applicable Special Packaging Instruction listed in the List of Special Packaging Instructions LSPI 8401840 for the level of protection specified in the contract (see 6.2.1).

5.2.2 Level X-industrial packaging. Repair parts preservation, packaging, and packing shall be in accordance with ASTM D 3951 when Level X protection is specified in the contract (see 6.2.1).

5.3 Marking.

5.3.1 Levels A and B. Marking of unit packages and exterior shipping containers shall be in accordance with Special Packaging Instruction P8401840.

5.3.2 Level X - Industrial. Marking shall be in accordance with ASTM D 3951.

5.3.3 Bar code marking. When specified, (see 6.2.1.k) bar code marking shall be in accordance MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended Use. The cannon and mount specified herein is intended to be used as a ground mounted mortar system.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.
b. Requirements for the first article to be submitted by the contractor (see 3.2).
c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.
d. All inspections and tests data that is required.
e. At whose expense a retest may be performed.
f. Serial numbers for the barrel (cannon) and mount assemblies.
g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3).
h. Availability of inspection equipment from the Government.
i. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment.
j. The levels of preservation, packaging, and packing required for cannon (see 5.1) and repair parts (see 5.2).
k. When applicable, bar code marking requirements (see 5.3).

6.2.2 Contract data requirements. Monthly reports of the results of final

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examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2.2.1 When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.1	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition funtions should be obtained from the Naval Publications and Forms Center or as directed by the Procuring Contracting Officer).

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-STD-45662.

6.5 Subject term (keyword) listing.

Cannon
Mortar
Mount

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR
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